Back

Time to Practice: Manifold Learning (Multidimensional scaling: Parts 1 & 2)
Graded Assignment • 15 min

Due Oct 14, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

1. What is the primary goal of manifold learning techniques in dimensionality reduction? 1 / 1 point

- To classify high-dimensional data into distinct categories.
- To project high-dimensional data onto a lower-dimensional space while preserving certain properties.
- To enhance the visualization of data by increasing the number of dimensions.
- To normalize the features of the original data

Nice work

Manifold learning techniques aim to reduce the dimensionality of complex data while maintaining properties such as distances or local neighborhood structures, making patterns more visible.

2. What is the primary goal of *non-metric* multidimensional scaling? 1 / 1 point

- To maximize the variance explained by the first few dimensions
- To predict the outcome variable from the high-dimensional data set
- To find a low-dimensional representation that preserves the ordinal relationships of the data.
- To cluster the data into distinct groups based on similarity

Nice work

Non-metric MDS aims to maintain the ordinal relationships, such as relative rankings or preferences, in a lower-dimensional space, rather than preserving exact distances.

3. What does traditional multidimensional scaling (MDS) primarily preserve in its dimensionality reduction process? 1 / 1 point

- Cluster memberships of points
- Individual point positions
- Local distances between points
- Global distances between points

Nice work

MDS focuses on preserving the global distances between points in the high-dimensional space when projecting them onto a lower-dimensional space.

4. Suppose you want to perform non-metric multidimensional scaling (MDS) by using a dissimilarity matrix as input, but due to dataset corruption some of the points have missing features, making it impossible to compute distances between some pairs of points. This causes missing values in the dissimilarity matrix. How do you handle this case so that it's still possible to use non-metric MDS? 1 / 1 point

- In the MDS objective function, set the weight of any pair of data points that involves missing data (has a missing dissimilarity entry) to 0
- Delete rows and columns with missing data
- Replace missing values with the mean of the matrix column containing the pair

Nice work

In cases where dissimilarity cannot be computed due to missing data, the weight of any pair involving such data points is set to 0 to effectively ignore these pairs in the analysis.

5. What is a key difference between a metric and a dissimilarity measure? 1 / 1 point

- Scale of measurement
- Computational complexity
- Symmetry requirement
- Type of data used

Nice work

A metric d(x, y) requires symmetry, meaning for two data points x and y, d(x, y) must equal d(y, x), whereas a dissimilarity measure does not necessarily have to be symmetric, so that d(x, y) may not equal d(y, x).

6. What is a key difference in the optimization goals of Principal Component Analysis (PCA) compared to Multidimensional Scaling (MDS)? 1 / 1 point

- PCA reduces dimensions, MDS reduces noise
- PCA minimizes distance distortion error, MDS maximizes between-points similarity
- PCA maximizes variance along the principal components, MDS aims to preserve distances
- PCA focuses on correlation, MDS focuses on clustering

Nice work

PCA aims to maximize the variance along the principal components, whereas MDS focuses on preserving the actual distances between data points in the dimensionality reduction process.